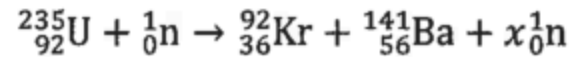
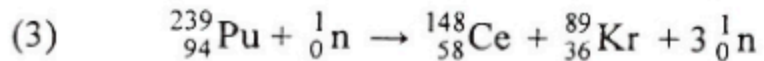
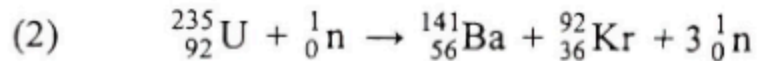
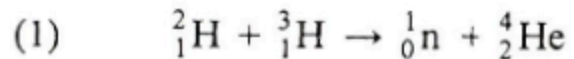


33. What is the value of  $x$  for the fission reaction below ?



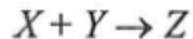
- A. 1
- B. 2
- C. 3
- D. 4

31. Which of the following nuclear reactions is/are possible for a chain reaction to occur ?



- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

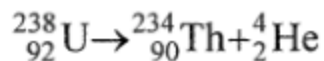
- \*33. In the following nuclear reaction, nuclide  $X$  and  $Y$  fuse to form nuclide  $Z$  under suitable conditions and the amount of energy released is 7.16 MeV.



The total mass of  $X$  and  $Y$  is

- A. 0.00385 u less than the mass of  $Z$ .
- B. 0.00385 u more than the mass of  $Z$ .
- C. 0.00769 u less than the mass of  $Z$ .
- D. 0.00769 u more than the mass of  $Z$ .

\*33. The following shows the decay of uranium-238 ( ${}^{238}_{92}\text{U}$ ).



Given that :      mass of  ${}^{238}_{92}\text{U} = 238.05079 \text{ u}$   
                         mass of  ${}^{234}_{90}\text{Th} = 234.04363 \text{ u}$   
                         mass of  ${}^4_2\text{He} = 4.00260 \text{ u}$

Which of the following statements is/are correct ?

- (1)      The temperature required to start the decay is about  $10^7 \text{ K}$ .
  - (2)      The energy released in the decay of one uranium-238 nucleus is  $4.25 \text{ MeV}$ .
  - (3)      All the energy released in the decay becomes the kinetic energy of  ${}^4_2\text{He}$ .
- 
- A.      (1) only
  - B.      (2) only
  - C.      (1) and (3) only
  - D.      (2) and (3) only

- \*33. Given: mass of a neutron =  $16749 \times 10^{-31}$  kg  
mass of a proton =  $16726 \times 10^{-31}$  kg  
mass of an electron =  $9 \times 10^{-31}$  kg

In a nuclear reaction, a neutron becomes a proton and a  $\beta$  particle. Estimate the energy released in this process.

- A. 1.8 MeV
- B. 1.3 MeV
- C. 0.79 MeV
- D. 0.51 MeV

\*36. The sun releases huge amount of energy through thermonuclear fusion while at the same time its mass decreases. The average power released by the sun is about  $3.8 \times 10^{26}$  W. Estimate the decrease in mass of the sun in one second.

- A.  $4.2 \times 10^6$  kg
- B.  $4.2 \times 10^9$  kg
- C.  $1.3 \times 10^{15}$  kg
- D.  $1.3 \times 10^{18}$  kg

- \*33. Given: mass of proton = 1.007276 u  
mass of neutron = 1.008665 u  
mass of  ${}^3_2\text{He}$  nucleus = 3.016030 u  
1 u = 931 MeV

When a  ${}^3_2\text{He}$  nucleus is formed from 2 protons and 1 neutron,

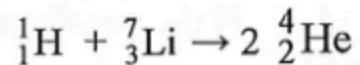
- A. 6.7 MeV energy is released.
- B. 6.7 MeV energy is required.
- C. 8.0 MeV energy is released.
- D. 8.0 MeV energy is required.

- \*33. A radium nucleus decays to a radon nucleus by emitting an  $\alpha$ -particle. The energy released in the process is 4.9 MeV. Compared to the mass of a radium nucleus, the total mass of a radon nucleus and an  $\alpha$ -particle is

- A.  $5.4 \times 10^{-11}$  kg less.
- B.  $5.4 \times 10^{-11}$  kg more.
- C.  $8.7 \times 10^{-30}$  kg less.
- D.  $8.7 \times 10^{-30}$  kg more.



- \*32. When a proton ( ${}^1_1\text{H}$ ) bombards a lithium nucleus ( ${}^7_3\text{Li}$ ), two  $\alpha$  particles ( ${}^4_2\text{He}$ ) are produced.



The energy released in this nuclear reaction is 17.32 MeV. Find the mass of a lithium nucleus.

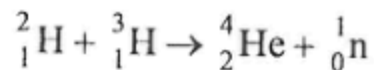
Given: mass of  ${}^1_1\text{H} = 1.0078 \text{ u}$   
mass of  ${}^4_2\text{He} = 4.0026 \text{ u}$

- A. 6.9788 u
- B. 6.9974 u
- C. 7.0017 u
- D. 7.0160 u

33. Which statement about nuclear fission is correct ?

- A. It involves two nuclei combined together.
- B. All nuclear fission reactions are spontaneous.
- C. Rate of fission reaction is temperature dependent.
- D. Fission of heavy nuclei yields products that are more stable in terms of energy.

- \*33. Consider the following nuclear reaction:



Using the data below, estimate the energy released when 5 g of  ${}^2_1\text{H}$  completely reacts with  ${}^3_1\text{H}$  through the reaction.

$$\text{mass of } {}^2_1\text{H} = 2.0136 \text{ u}$$

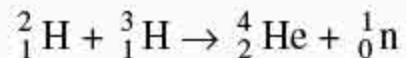
$$\text{mass of } {}^3_1\text{H} = 3.0155 \text{ u}$$

$$\text{mass of } {}^4_2\text{He} = 4.0015 \text{ u}$$

$$\text{mass of } {}^1_0\text{n} = 1.0087 \text{ u}$$

- A.  $3 \times 10^{-12} \text{ J}$
- B.  $3 \times 10^{-10} \text{ J}$
- C.  $4 \times 10^{12} \text{ J}$
- D.  $2 \times 10^{18} \text{ J}$

- \*36. For the following nuclear reaction, state the type of reaction and determine the energy released.



Given: mass of  ${}^2_1\text{H} = 2.014 \text{ u}$

mass of  ${}^3_1\text{H} = 3.016 \text{ u}$

mass of  ${}^4_2\text{He} = 4.003 \text{ u}$

mass of  ${}^1_0\text{n} = 1.009 \text{ u}$

|    | Type of reaction | Energy released |
|----|------------------|-----------------|
| A. | fusion           | 0.018 MeV       |
| B. | fusion           | 16.76 MeV       |
| C. | fission          | 0.018 MeV       |
| D. | fission          | 16.76 MeV       |

34. Which of these is a nuclear fusion reaction ?

